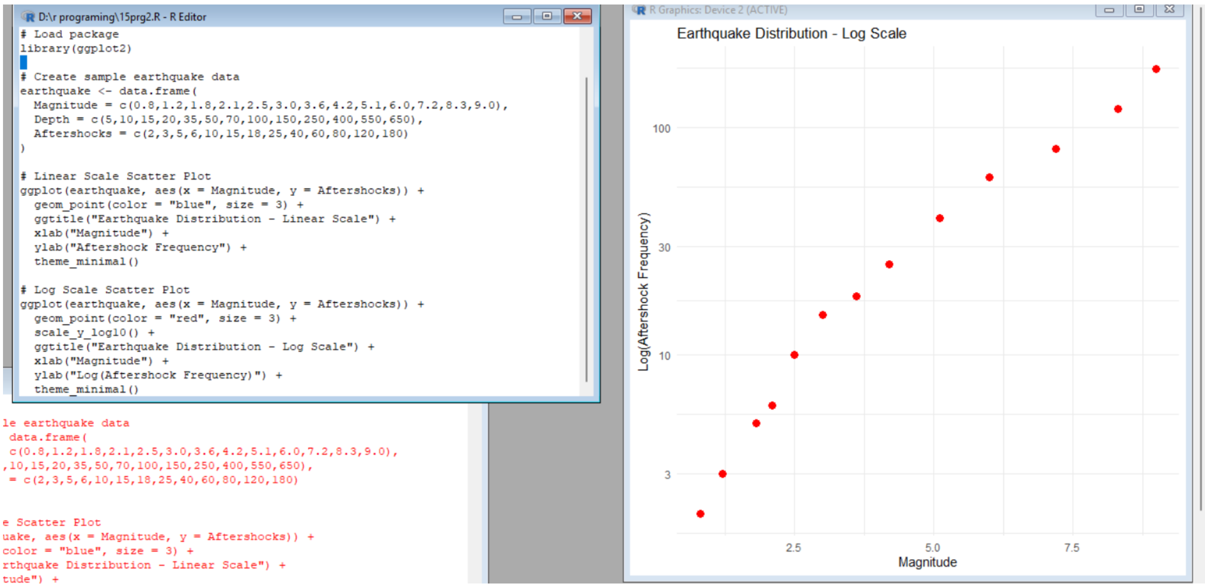
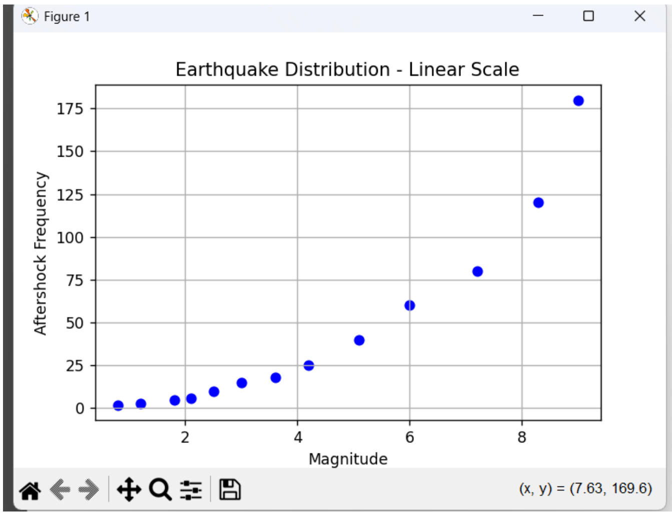
2.**Earthquake Risk Analysis**

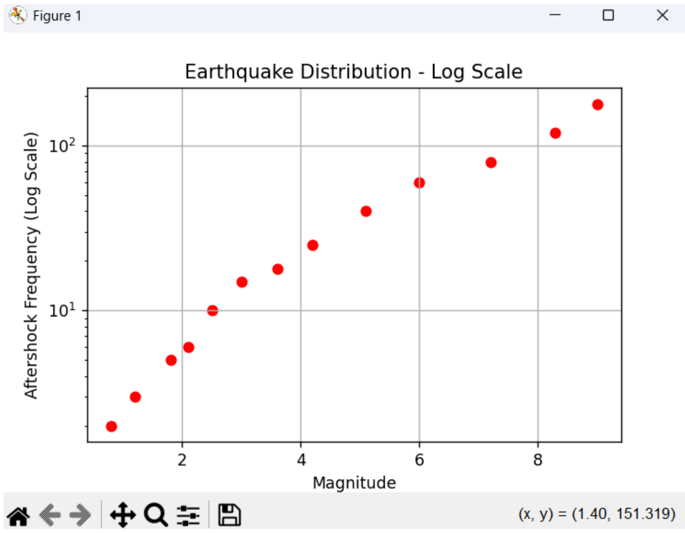
R-code

```
# Install package (Run once)
install.packages("ggplot2")
# Load package
library(ggplot2)
# Create sample earthquake data
earthquake <- data.frame(
  Magnitude = c(0.8,1.2,1.8,2.1,2.5,3.0,3.6,4.2,5.1,6.0,7.2,8.3,9.0),
  Depth = c(5,10,15,20,35,50,70,100,150,250,400,550,650),
  Aftershocks = c(2,3,5,6,10,15,18,25,40,60,80,120,180)
)
# Linear Scale Scatter Plot
ggplot(earthquake, aes(x = Magnitude, y = Aftershocks)) +
  geom_point(color = "blue", size = 3) +
  ggtitle("Earthquake Distribution - Linear Scale") +
  xlab("Magnitude") +
  ylab("Aftershock Frequency") +
  theme_minimal(
# Log Scale Scatter Plot
ggplot(earthquake, aes(x = Magnitude, y = Aftershocks)) +
  geom_point(color = "red", size = 3) +
  scale_y_log10() +
  ggtitle("Earthquake Distribution - Log Scale") +
  xlab("Magnitude") +
  ylab("Log(Aftershock Frequency)") +
  theme_minimal()
```



Python code





3.**Wind Direction Analysis for Airport Operations**

R-code

Install packages (Run only once)

install.packages("ggplot2"

Load library

library(ggplot2)

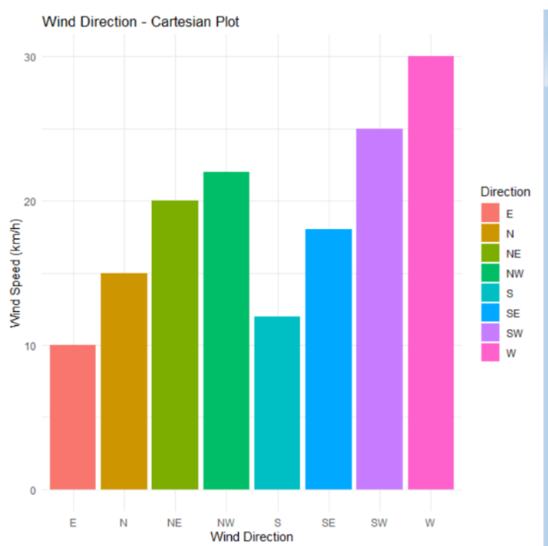
wind <- data.frame(

Direction = c("N","NE","E","SE","S","SW","W","NW"),

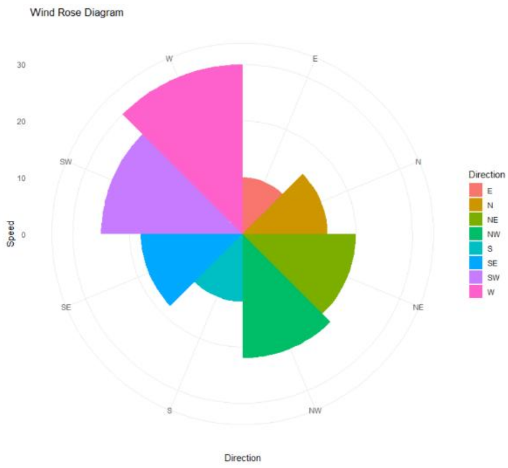
Speed = c(15,20,10,18,12,25,30,22)

)

ggplot(wind, aes(x = Direction, y = Speed, fill = Direction)) +

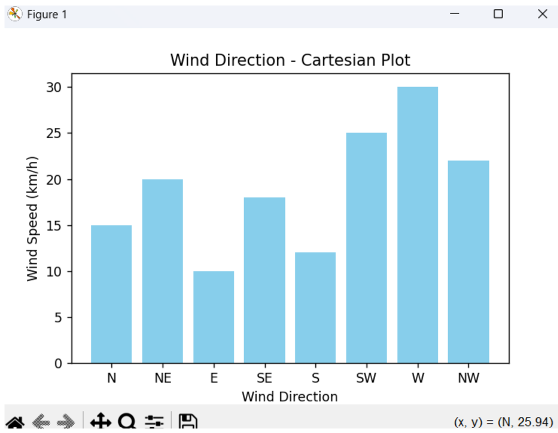


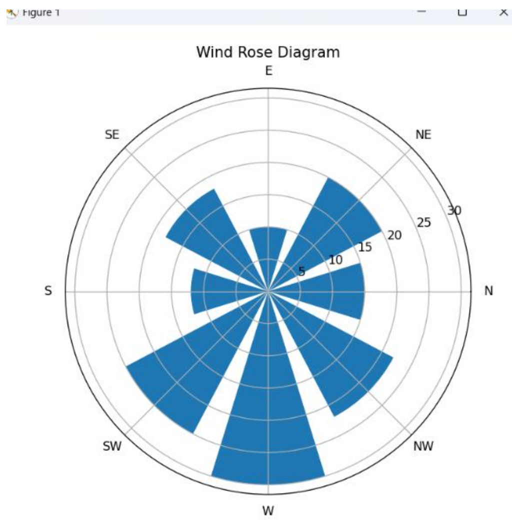
```
  geom_bar(stat = "identity") +
  labs(title = "Wind Direction - Cartesian Plot",
       x = "Wind Direction",
       y = "Wind Speed (km/h)") +
  theme_minimal()
# Wind Rose (Polar Plot)
ggplot(wind, aes(x = Direction, y = Speed, fill = Direction)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar() +
  labs(title = "Wind Rose Diagram") +
  theme_minimal()
```



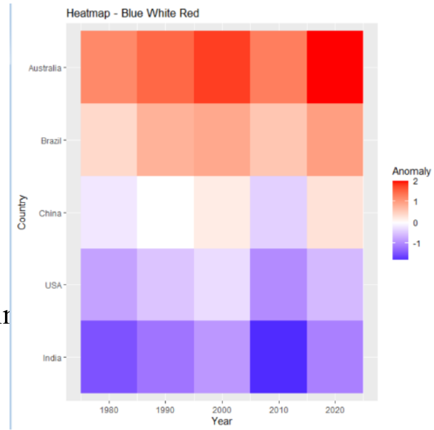
python -code

```
import matplotlib.pyplot as plt
import numpy as np
directions = ['N','NE','E','SE','S','SW','W','NW']
speed = [15,20,10,18,12,25,30,22
plt.figure(figsize=(6,4))
plt.bar(directions, speed, color='skyblue')
plt.title("Wind Direction - Cartesian Plot")
plt.xlabel("Wind Direction")
plt.ylabel("Wind Speed (km/h)")
plt.show(
angles = np.linspace(0, 2*np.pi, len(speed), endpoint=False
fig = plt.figure(figsize=(6,6))
ax = fig.add_subplot(111, polar=True)
ax.bar(angles, speed, width=0.6)
ax.set_xticks(angles)
ax.set_xticklabels(directions)
ax.set_title("Wind Rose Diagram")
plt.show()
```



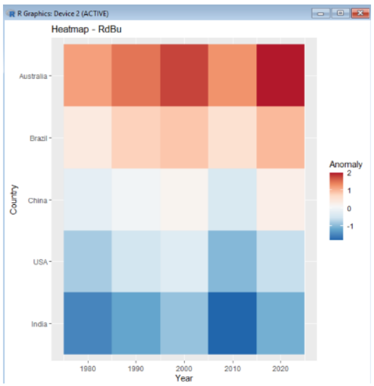


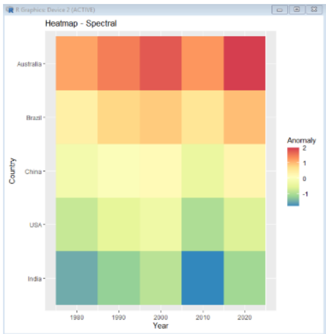
5. **Global Climate Change Study**



R-code

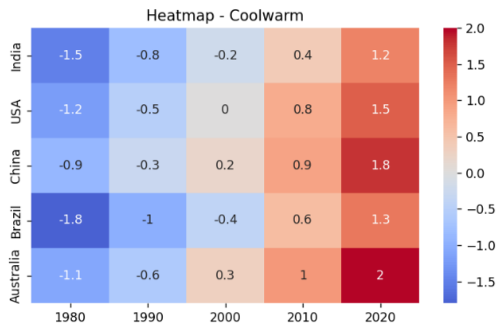
```
# Palette 1: Blue-White-Red
ggplot(climate, aes(Year, Country, fill = Anomaly)) +
  geom_tile() +
  scale_fill_gradient2(low="blue", mid="white", high="red", midpoi
  ggtitle("Heatmap - Blue White Red")
# Palette 2: RdBu
ggplot(climate, aes(Year, Country, fill = Anomaly)) +
  geom_tile() +
  scale_fill_distiller(palette="RdBu", direction=-1) +
  ggtitle("Heatmap - RdBu")
# Palette 3: Spectral
ggplot(climate, aes(Year, Country, fill = Anomaly)) +
  geom_tile() +
  scale_fill_distiller(palette="Spectral", direction=-1) +
  ggtitle("Heatmap - Spectral")
```



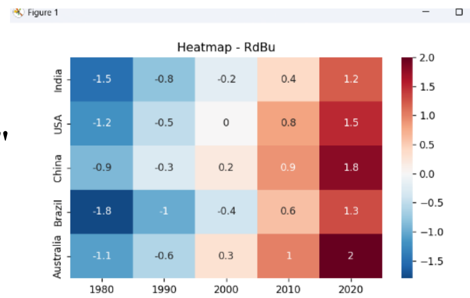


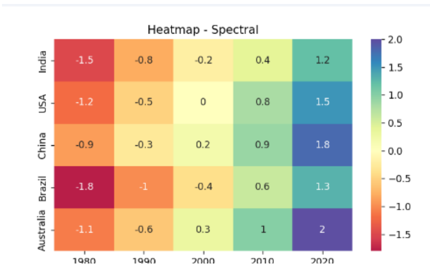
python -cod

```
# Palette 1
plt.figure(figsize=(7,4))
sns.heatmap(data, cmap="coolwarm", center=0, annot=True)
plt.title("Heatmap - Coolwarm")
plt.show()
```




```
# Palette 2
plt.figure(figsize=(7,4))
sns.heatmap(data, cmap="RdBu_r"
plt.title("Heatmap - RdBu")
plt.show()
```





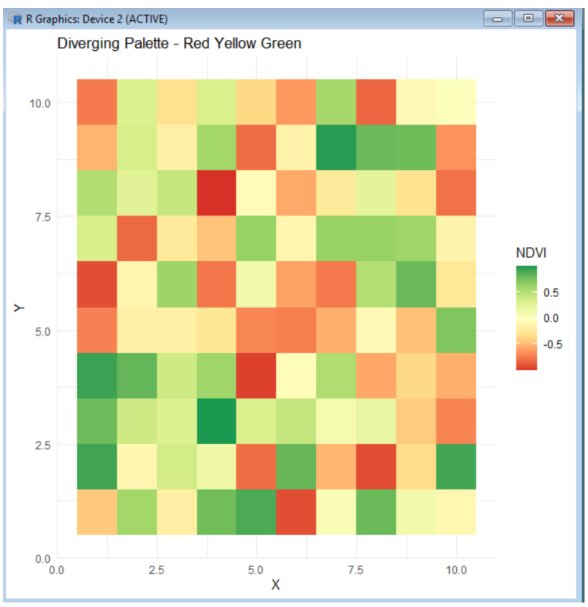

```
# Palette 3
plt.figure(figsize=(7,4))
sns.heatmap(data, cmap="Spectral", center=0, annot=True)
```

```
plt.title("Heatmap - Spectral")
plt.show()
```

7. Satellite Image Vegetation Analysis

R code

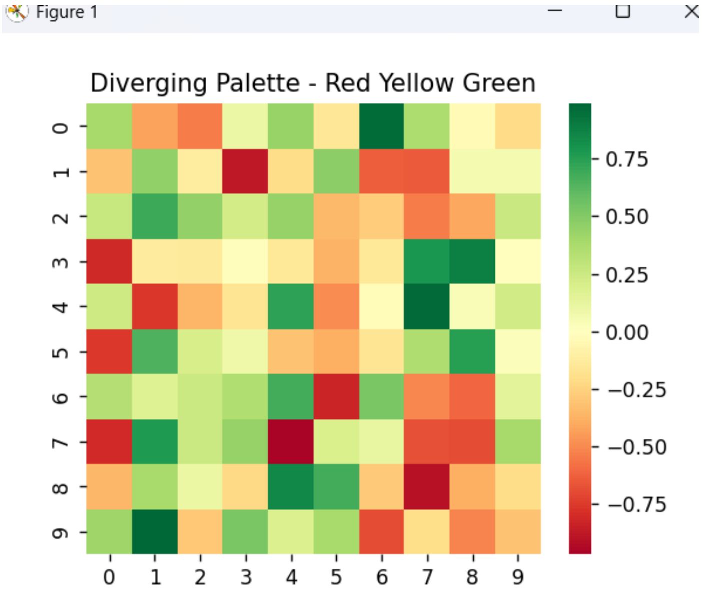
```
# Palette 1: Greens (Sequential)
ggplot(ndvi, aes(X, Y, fill = NDVI)) +
  geom_tile() +
  scale_fill_distiller(palette = "Greens") +
  ggtitle("Sequential Palette - Greens") +
  theme_minimal()
# Palette 2: YlGn (Sequential)
ggplot(ndvi, aes(X, Y, fill = NDVI)) +
  geom_tile() +
  scale_fill_distiller(palette = "YlGn") +
  ggtitle("Sequential Palette - Yellow-Green") +
  theme_minimal()
# Palette 3: RdYlGn (Diverging)
ggplot(ndvi, aes(X, Y, fill = NDVI)) +
  geom_tile() +
  scale_fill_distiller(
    palette = "RdYlGn",
    direction = 1
  ) +
  ggtitle("Diverging Palette - Red Yellow Green") +
```



python code

 # Diverging Palette - RdYlGn

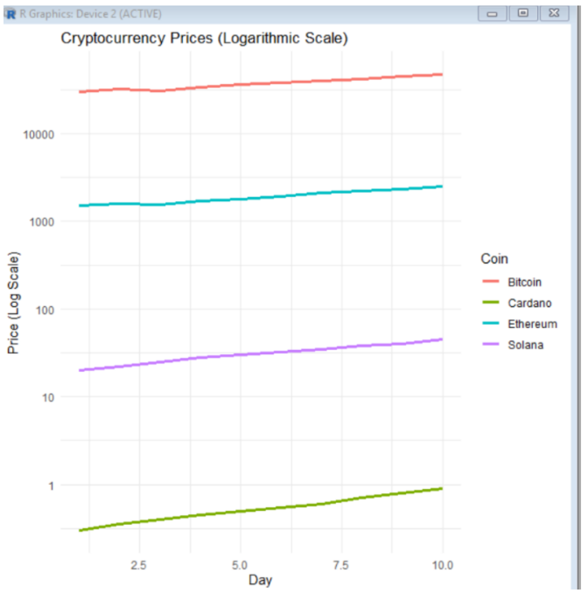
```
plt.figure(figsize=(5,4))
sns.heatmap(ndvi, cmap="RdYlGn", center=0)
plt.title("Diverging Palette - Red Yellow Green")
plt.show()
```



8. Cryptocurrency Market Volatility

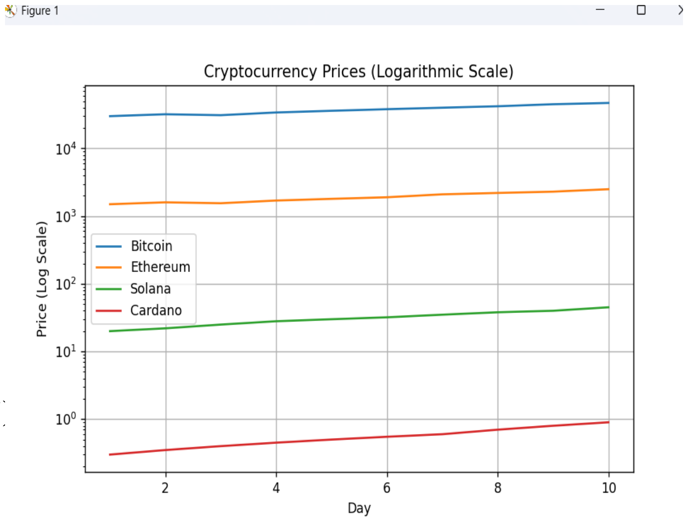
R-code

```
# Logarithmic Scale
ggplot(crypto, aes(x = Day, y = Price, color = Coin)) +
  geom_line(size = 1.2) +
  scale_y_log10() +
  labs(
    title = "Cryptocurrency Prices (Logarithmic Scale)",
    x = "Day",
    y = "Price (Log Scale)"
  ) +
  theme_minimal()
```



python code

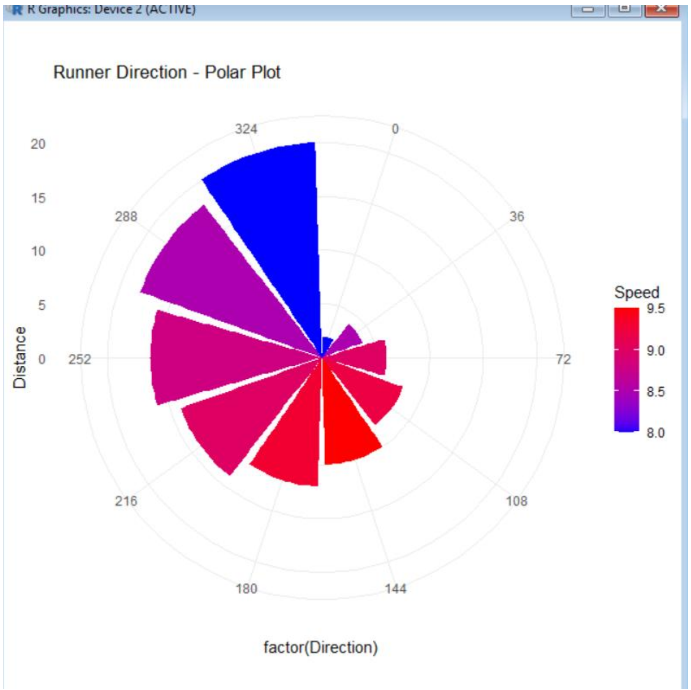
```python
# Logarithmic Scale
plt.figure(figsize=(8,5))
plt.plot(days, bitcoin, label="Bitcoin")
plt.plot(days, ethereum, label="Ethereum")
plt.plot(days, solana, label="Solana")
plt.plot(days, cardano, label="Cardano")
plt.yscale("log")
plt.title("Cryptocurrency Prices (Logarithmic Scale)"
plt.xlabel("Day")
plt.ylabel("Price (Log Scale)")
plt.legend()
plt.grid(True)
plt.show()
```



9. **Marathon Runner Performance Analysis**

R-code

```r
# Polar Coordinate Plot
ggplot(runner,
       aes(x=factor(Direction),
           y=Distance,
           fill=Speed)) +
  geom_bar(stat="identity") +
  coord_polar() +
  scale_fill_gradient(low="blue", high="red") +
  labs(title="Runner Direction - Polar Plot") +
  theme_minimal()
```



Python code

```
# Polar Plot
plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
bars = ax.bar(direction, distance, width=0.35,
        color=plt.cm.coolwarm(np.array(speed)/max(speed)))
ax.set_title("Runner Direction - Polar Plot")
plt.show()
```